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Operational Risk Dictionary

Pipeline Exception & Criticality Framework

This dictionary maps operational vulnerabilities tracked across our multi-jurisdictional infrastructure. Severity tiers dictate automated pipeline triggers within `report_generation.py` and escalation priorities for `email_sending.py`.

Risk Category/Identifier					
	≡ Sector	Risk ≡ Category/Identifier	≡ Operational Hazard Definition	≡ Script Failure Trigger	≡ Core Mitigation Control
1	Plastic	Environmental Risk (ERA 12)	Loss of microplastic pellet containment causing storm-water pollution.	Non-compliance with Operation Clean Sweep standards.	Daily machine safety gate isolation validation.
2	Food	Biosecurity & Trade Waste Liquid Risk	Fats, oils, and grease structural overflows breaching municipal sewer lines.	Missing or delayed Grease Trap maintenance records.	Mandatory Food Safety Supervisor credential checks.
3	Furniture	Combustible Dust Control	Accumulation of explosive particulate matters in high-density manufacturing areas.	Local Exhaust Ventilation (LEV) pressure drop failures.	Regular LEV automated particle validation assays.
4	Wooden	Chemical Timber Treatment	Soil or groundwater contamination from chemical preservative tracking.	Deficient bunding or containment structure walls.	Scheduled soil baseline sample validation tests.
5	Clothes	High-Density Logistics Risk	Blocked escape infrastructure paths in major regional distribution centres.	Failed warehouse stockroom walkway clearing validation.	Routine fire egress tag and route clearing checks.
6	Groceries	Commercial Organic Waste / Cold Chain	Growth of dangerous foodborne pathogens due to temperature variations.	Cold chain logs reading outside optimal baseline constraints.	Automated fridge sensor calibration logs.

Risk Scoring Matrix & Automated Script Logic

Tier 1: Critical Operational Risk

- **Definition:** Severe breach of statutory local government bylaws or state environmental acts (e.g., Plastic processing spill or Food biosecurity failure).
- **Pipeline Action:** Automated `report_generation.py` script isolates these records into distinct regional PDF books, prompting an immediate high-priority mail-relay dispatch through `email_sending.py` to local council auditors.

Tier 2: Moderate Operational Risk

- **Definition:** Minor data logging gaps or tracking delays (e.g., missing a scheduled fire extinguisher tag check or a routine waste sorting count).
- **Pipeline Action:** Flagged inside the master loop and catalogued inside the Commute & Site Log for review during the upcoming multi-jurisdictional mapping cycle.

Systems Architecture Alignment

This dictionary is hardcoded into our data validation schemas. The `for category in categories:` loop inside our core data engine relies on these precise risk definitions to parse out empty rows, sanitize whitespace bugs, and structure data uniformly across all 218 localized tracking records.